Problem Statement

2Market, a global supermarket chain operating online and in-store, seeks to better understand its diverse customer base by analysing demographics like age, location, and shopping habits. By gaining insights into these groups, 2Market aims to strengthen marketing strategies, and improve the overall shopping experience. The resulting insights will guide smarter decisions in advertising, product placement, and promotions.

Questions for the 2Market Team

- What customer demographic data will be available?
- What current strategies are in place, and how are they performing?

Questions About the Data

- Are there specific success metrics beyond purchases?
- Do performance ratings vary by season or region?

Analytical approach

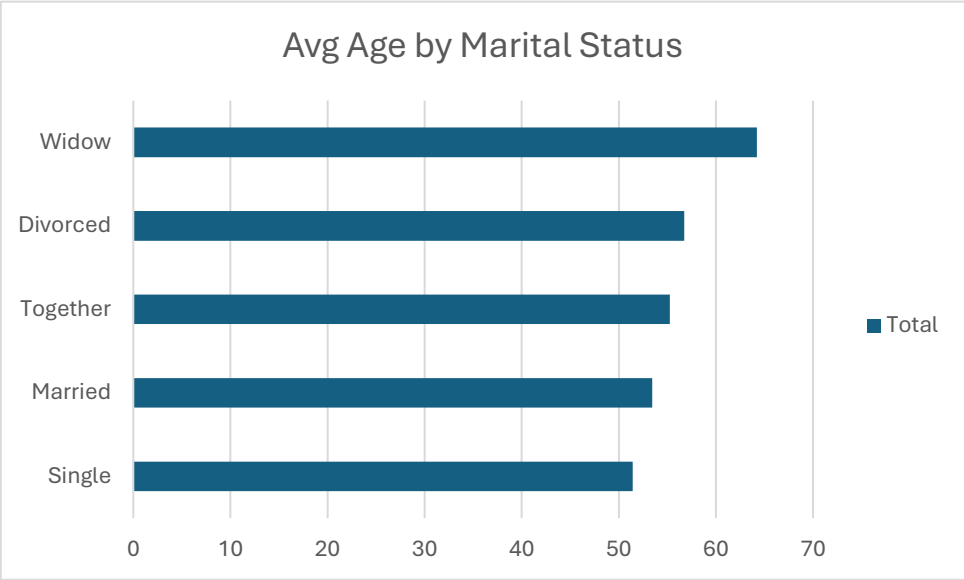
The first step I took was reading the metadata_2Market.txt file to identify the provided data and begin planning my assessment. I then imported 'marketing_data.csv' into Excel, checking for blank spaces, spelling errors, and data types. Dates in 'Dt_Customer' were formatted as TEXT, so I used 'Text to Columns' to convert them to MDY. The 'Income' column was also TEXT; I removed the '$' symbol using 'SPLIT' and ensured the values were numeric.

No duplicates were found in the 'Id' column. I saved the cleaned CSV as an XML file for multi-tab analysis. I found abnormal records in 'Marital_Status' ('Absurd', 'YOLO', etc.) and 'Education' ('2n Cycle'). I chose not to remove them due to the volume and will filter results accordingly.

To explore the relationship between age and marital status, I calculated customer ages using =YEAR(TODAY()) - C2. I also found outliers, such as customers over 100, with possible errors in marital status and education records.

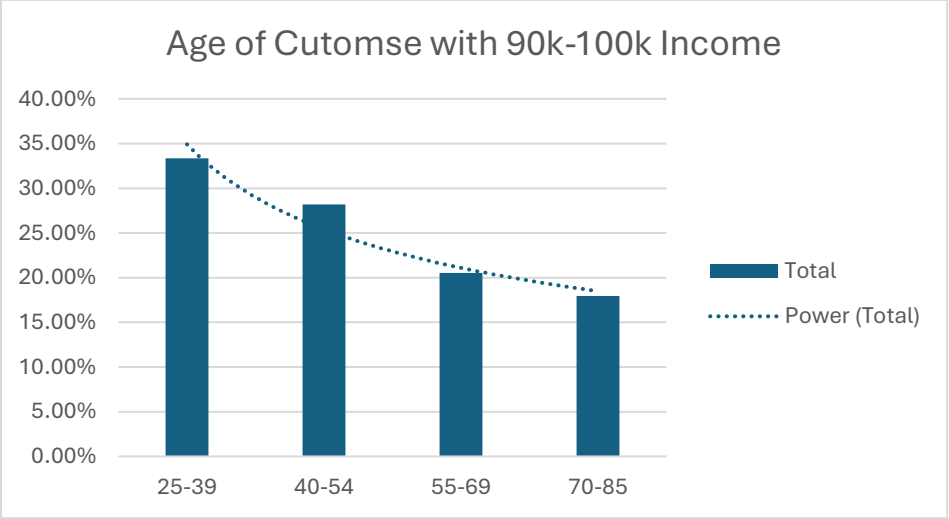
After this, I used the Pivot Table functionality to display a table with the average age by marital status, as shown below:

Row Labels	Average of age
Single	51
Married	53
Together	55
Divorced	57
Widow	64
Grand Total	**54**



Looking at the distribution of customers with an income between $90,000 and $100,000, there are none over the age of 100. The distribution is balanced across the age bins, but the count decreases as age increases:

Row Labels	Count of Income
25-39	33.33%
40-54	28.21%
55-69	20.51%
70-85	17.95%
Grand Total	**100.00%**



I continued my analysis in SQL after uploading both CSV files into the relevant tables. My findings are as follows:

- Spain has the highest total spend at $659,557, determined using the SUM function.
- Alcohol is the most popular product in every country.

	country character varying (10)	total_alcohol bigint	total_vegetables bigint	total_meat bigint	total_fish bigint	total_chocolates bigint	total_commodities bigint	most_popular_product text
1	SP	336392	28288	178409	40153	30134	46181	Alcohol
2	SA	105918	8937	58398	13670	9019	15129	Alcohol
3	CA	82950	7599	45296	9754	7583	11858	Alcohol
4	AUS	42281	3587	22203	5334	4068	6887	Alcohol
5	GER	36776	2980	20272	4601	2801	5768	Alcohol
6	IND	35914	3785	23679	4814	3218	5972	Alcohol
7	US	32214	3034	20185	4411	2863	4839	Alcohol
8	ME	1729	8	817	226	122	220	Alcohol

- Alcohol is the most popular product across each marital status.

	marital_status character varying (50)	total_alcohol bigint	total_vegetables bigint	total_meat bigint	total_fish bigint	total_chocolates bigint	total_commodities bigint	most_popular_product text
1	Married	256976	21981	137888	30395	22926	36719	Alcohol
2	Together	176715	14612	95374	22383	15031	24754	Alcohol
3	Single	137217	12840	87064	18262	12751	20397	Alcohol
4	Divorced	75364	6363	34848	8130	6222	10739	Alcohol
5	Widow	27902	2422	14085	3793	2878	4245	Alcohol

- Alcohol is also the most popular product among customers with children or teens.

	has_children_or_teens text	total_alcohol bigint	total_vegetables bigint	total_meat bigint	total_fish bigint	total_chocolates bigint	total_commodities bigint	most_popular_product text
1	Has Children/Teens	365935	25297	135126	34874	26206	56601	Alcohol
2	No Children/Teens	308239	32921	234133	48089	33602	40253	Alcohol

Using SQL, I delved into the 'ad_data'. My findings include:

- The most effective social media platform for advertising in each country

	country character varying (10)	twitter_success bigint	instagram_success bigint	facebook_success bigint	best_platform text
1	AUS	13	23	15	Instagram
2	CA	43	47	41	Instagram
3	GER	19	14	13	Twitter
4	IND	14	12	13	Twitter
5	ME	0	0	0	Twitter
6	SA	34	42	35	Instagram
7	SP	159	182	163	Instagram
8	US	8	10	12	Facebook

- The most effective social media platform by marital status

	marital_status character varying (50)	twitter_success bigint	instagram_success bigint	facebook_success bigint	best_platform text
1	Single	57	63	62	Instagram
2	Together	80	89	67	Instagram
3	Absurd	0	2	2	Instagram
4	Married	103	129	123	Instagram
5	Divorced	33	31	29	Twitter
6	YOLO	0	0	0	Twitter
7	Widow	17	16	9	Twitter
8	Alone	0	0	0	Twitter

- The most effective platform by lead conversion

	country character varying (10)	total_alcohol bigint	total_vegetables bigint	total_meat bigint	total_fish bigint	total_chocolates bigint	total_commodities bigint	twitter_success bigint	instagram_success bigint	facebook_success bigint	most_effective_platform text
1	AUS	42752	3689	22328	5546	4129	7132	13	23	15	Instagram
2	CA	84066	7681	45925	9980	7607	12144	43	47	41	Instagram
3	GER	36776	2980	20272	4601	2801	5768	19	14	13	Twitter
4	IND	36236	3788	23729	4818	3221	6014	14	12	13	Twitter
5	ME	1729	8	817	226	122	220	0	0	0	Twitter
6	SA	105918	8937	58398	13670	9019	15129	34	42	35	Instagram
7	SP	336392	28288	178409	40153	30134	46181	159	182	163	Instagram
8	US	32214	3034	20185	4411	2863	4839	8	10	12	Facebook

Dashboard Design

Before making the dashboard in Tableau, I imported the marketing_data_cleaned.csv. I then left-joined the ad_data.csv, matching the ID columns in each. My first aim here is to create a visualization to compare sales per country. This uses the location of the countries on record, along with the spend per country. The table on top shows the spend per marital status, per age group, per country. Using filters, you can drill down into each country to see which products sell best in the best locations, and more, by demographic. Finally, on this dashboard we have a barchart showing the average spend per age group. To allow us to assess which age groups are the biggest spenders.

The next visualization I've made provides insight into the social media marketing effectiveness. This dashboard allows you to manipulate the filters to delve into effectiveness by country, marital status, and age. This dashboard contains three tables to achieve this. In the process of making these tables, I found that there was no direct way to compare or find the standout social media platform. So, I made a parameter called 'Best Ad Platform' and created the following formula:

IF [Twitter ad] >= [Instagram ad] AND [Twitter ad] >= [Facebook ad] THEN "Twitter"
ELSEIF [Instagram ad] >= [Twitter ad] AND [Instagram ad] >= [Facebook ad] THEN "Instagram"
ELSEIF [Facebook ad] >= [Twitter ad] AND [Facebook ad] >= [Instagram ad] THEN "Facebook"
END

The next dashboard I created focuses on the income of 2Market customers. The first table shows the disparity between 2Market customers who make $90,000 - $100,000 a year, and the brackets on either side, in the form of a pie chart. Below is a table that shows the average spend of bins in 15-year splits. The column on the right shows the spend by income in the form of a scatter graph, accompanied by a trend line for better understanding. This entire dashboard can be filtered by marital status, country, education, and income. There is also a key for the colours represented in the dashboard.

Insights

From the demographic analysis, the average age of 2Market's customers is 54. Singles represent the youngest group with an average age of 51, while Widows are the oldest, with an average age of 64. The mode age is 47, indicating that most of their customers fall into this age range, as visualized in the pivot tables and dashboards.

In terms of spending habits, Spain stands out with a total spend of $659,557. This suggests that 2Market has a very loyal customer base in Spain, making it a key focus for growth. I'd recommend investigating competitors in this region to identify any market gaps. Additionally, we should explore what makes 2Market successful here and consider implementing these strategies in other regions.

Analysis across several tables highlights alcohol as the most popular product across all countries, regardless of marital status or household composition. This indicates strong demand and suggests that alcohol is a significant revenue driver for 2Market. Expanding the availability of this product could further increase sales.

Looking at the marketing, the strongest marketing out is Instagram, and this is evident in Spain. If we are looking to replicate the success achieved in other countries, you should look to copy the trends in lower performing countries like Montenegro.

Findings

During this

I found that the average age for 2Market's customers is 54. Examining the average ages across different marital statuses, I discovered that the youngest average age is for 'Singles' at 51, and the oldest average age is for 'Widowed' at 64. These results appear to be a natural outcome, with 'Married' having an average age of 53, 'Divorced' at 57, and 'Together' at 55.